

Train Your Own Summarizer: Simple Guide

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What you're doing

You're teaching a small AI model to summarize news articles. You take a ready-made model (Qwen3-4B), show it 10,000 news articles with their summaries, and it learns to write summaries in the same style. It runs on your own 12GB GPU and takes a few hours.

What you need

- ☐ An NVIDIA GPU with 12GB memory
- ☐ Linux, or Windows with WSL2 (Ubuntu)
- ☐ Python 3.10, 3.11 or 3.12
- ☐ About 30GB of free disk space
- ☐ The project files from the zip, unzipped into a folder called `summarizer`

The 5 steps

Open a terminal inside the `summarizer` folder and run these in order.

1. Install

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

This creates a clean Python space and installs everything. It takes 5-10 minutes.

2. Do a quick test run

```
python train.py --max_steps 30 --train_size 500
```

This trains for just a few minutes to make sure everything works. If it finishes without errors, move on.

3. Train for real

```
python train.py
```

This takes about 2–5 hours. You'll see a "loss" number printed every few seconds; it should go down over time. When it's done, your trained model is saved in the `outputs` folder.

4. Try it

```
python summarize.py --sample
```

This picks a random news article, shows your model's summary, and shows the real summary so you can compare. To summarize your own article, save it as a text file and run:

```
python summarize.py --file my_article.txt
```

5. Check the score

```
python evaluate_rouge.py --model base
python evaluate_rouge.py --model finetuned
```

This scores the original model and your trained model on 500 articles. A higher ROUGE number means summaries closer to the real ones. Your trained model should score higher than the base model.

If something goes wrong

Problem	Fix
"CUDA out of memory"	Close other apps using the GPU, then run <code>python train.py --batch_size 1 --grad_accum 16</code>
Training is extremely slow	Your GPU isn't being used. Run <code>nvidia-smi</code> to check it's detected, and reinstall if needed.
Install errors	Start fresh: delete the <code>.venv</code> folder and repeat Step 1
Training stopped halfway	Run <code>python train.py --resume</code> to continue where it left off